Analysis of parameter variations in a two-dimensional Von Kármán vortex street induced by a steady obstacle

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Abstract

Here goes the abstract of the paper. It should include the objective, method, main results, and conclusion in a compact form. The computational technique used to achieve the results of this

paper is to combine the use of the open-source software Basilisk, the Finite Volume Method (FVM) and the adaptive mesh refinement (AMR).

1 Introduction

The main purpose of this study is to model and analyse different aspects of a well known phenomenon in fluid dynamics: A Von Kármán Vortex Street, caused by an incoming flow over a steady obstacle. Firstly, the goal is to establish two baseline models, one with a laminar incoming flow and the other with a turbulent one. Secondly, various parameters that define this model are altered, in order to study how these changes affect each type of flow. Additionally to these objectives, it is aim to compare how these variations will evolve within the framework of the same type of baseline simulation. The parameters chosen to vary are: The Reynolds Number 1, the morphology of the object and the viscosity **??**. With all the combinations of changes a systematic sensitivity analysis is done.

After some research is easy to conclude that there are several studies about this type of vortex. This is, because it appears naturally in the contexts that are related with turbulent flows, which are present in almost every nature's fluid with any states of matter and geometry. The majority of the computational models done on this phenomenon concerned only about one feature of the fluid, and they make a deep study about that aspect. Such as..

On the other hand, the goal of this article is to make a contribution by analysing the effects of varying different characteristics of a certain fluid at the same time. Putting particular emphasis on how the combinations of these changes shifts the fluid behaviour.

1.1 Theoretical framework

The periodic shedding of vortices in the wake of a bluff body immersed in a viscous flow it is universally known as the *Von Kármán vortex street*, which stands as one of the most canonical and pedagogically rich phenomenon in fluid mechanics. First described analytically by Theodore von Kármán in 1911 [1], and embedded within the broader mathematical framework laid out by Lamb in his seminal treatise [2]. To understand this particular mechanism it is crucial to define the Reynolds Number, which will be widely used in the rest of the paper. This a particular number that is define based on some basic variables of the fluids, and is very powerful and useful because it represents the general behaviour of them.

$$Re = \frac{\rho U L}{\mu} = \frac{U L}{\nu} \tag{1}$$

The phenomenon emerges from an intrinsic hydrodynamic instability: when the Reynolds number exceeds a critical threshold (approximately $Re \gtrsim 47$ for a circular cylinder), the symmetric recirculation region attached to the obstacle becomes unstable, and the flow bifurcates into a time-periodic asymmetric regime characterized by two staggered rows of counter-rotating vortices that are periodically shed and convected downstream.

The governing equations are the Navier–Stokes equations, which in Cartesian coordinates (x, y, z) are represented by:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\,\nabla^2\mathbf{u} + \mathbf{f}, \tag{2}$$

That in this context is also use with the divergence-free constraint:

$$\nabla \cdot \mathbf{u} = 0, \tag{3}$$

where $\mathbf{u} = (u,\, v,\, w)^\top$ is the velocity field, p the mechanical pressure, ρ the (constant) fluid density, μ the dynamic viscosity, and $\mathbf{f}$ any external body force. Equation (2) expresses Newton's second law for a fluid parcel: the left-hand side is the material acceleration (local plus convective inertia), balanced on the right by the pressure gradient, viscous diffusion, and external forcing. In the two-dimensional reduction adopted here, $w \equiv 0$ and all z-derivatives vanish, yielding a system in (u, v, p) that retains the full nonlinear coupling responsible for the instability.

The boundary layer that develops along the obstacle surface plays a decisive role in vortex shedding. At low Re the layer remains attached and the wake is steady; as Re increases, the adverse pressure gradient on the lee side of the body forces flow separation, generating two shear layers that roll up alternately into discrete vortices. The non-dimensional frequency of this shedding process is quantified by the Strouhal number $St = fD/U$, where f is the vortex-shedding frequency and D is the characteristic diameter of the obstacle. For a circular cylinder in the laminar shedding regime $(47 \lesssim Re \lesssim 180)$ experiments consistently report $St \approx 0.2$ [2], a value that the present simulations are expected to reproduce and that serves as a primary validation metric. The transition to a turbulent inflow modifies this picture significantly: fluctuations in the incoming stream alter the separation point, broaden the spectral content of the wake, and shift the effective Re at which the Kelvin–Helmholtz instability in the shear layers becomes active, all of which are explored in the sensitivity analysis described in subsequent sections.

2 Methodology

General introduction of the numerical method, the adaptive mesh and the use of Basilisk and its features.

All simulations are performed with Basilisk, which is an open source software with a finite-volume solver on adaptive Cartesian meshes. Whose mesh-refinement capabilities make it especially well suited for resolving the thin boundary layer at the obstacle surface and the coherent vortical structures in the near wake.

The first step is to create two baseline models, one with a laminar incoming flow and the other with a turbulent one. After that, a systematic sensitivity analysis it is made, in which the obstacle's morphology, the Reynolds Number , and the dynamic viscosity μ are varied independently for both regimes.

Note that, even though, it can be argued that the density is a variable omitted in all the models that are previously explained, it is not. Due to, its relation with the viscosity and the Reynolds Number **??** D / $\mu Re = \rho UD/\mu$ensity changes in every simulation.

2.1 Finite Volume Method (FVM)

- *Discretization Schemes: Explicitly state the schemes used for the spatial derivatives (e.g., second-order central differencing for diffusion terms, QUICK or upwind schemes for convective terms).*
- *Time Integration: Detail how you advance the simulation in time (e.g., implicit Euler, fractional-step methods, or a specific Runge-Kutta scheme).*
- *Linear Solvers: FVM ultimately reduces your fluid problem to massive systems of linear equations. You must state how these are solved. If you implemented a custom numerical system to solve these equations—such as utilizing P, L, and U matrix decomposition—detail this algorithm and its convergence criteria.*
- *Boundary Conditions: Specify the exact numerical treatment at the inlet, outlet, and the surface of the bluff body generating the wake.*

Now talking about the numerical method, the finite volume method (FVM) is a spatial discretization technique in which the computational domain is partitioned into a finite number of non-overlapping control volumes Ω_i, and the governing equations are enforced in their *integral* form over each cell. Integrating equation (2) over an arbitrary control volume Ω_i and applying the divergence theorem, the volume integral of any flux divergence is converted into a surface integral over the cell boundary $\partial\Omega_i$, yielding:

$$\frac{\partial}{\partial t}\int_{\Omega_i} \rho\,\mathbf{u}\,dV + \oint_{\partial\Omega_i} \rho\,\mathbf{u}\,(\mathbf{u}\cdot\hat{n})\,dS = -\oint_{\partial\Omega_i} p\,\hat{n}\,dS + \oint_{\partial\Omega_i} \mu\,(\nabla\mathbf{u}\cdot\hat{n})\,dS, \tag{4}$$

Where $\hat{n}$ is the outward unit normal to $\partial\Omega_i$. The key numerical step consists in approximating each surface integral by the sum of fluxes evaluated at the faces shared between adjacent cells; in this way, a flux leaving one control volume enters its neighbor exactly, making the method *locally and globally conservative* by construction—a property that finite difference discretizations do not inherently guarantee. The volume integral on the left is approximated by the cell average $\rho\,\bar{\mathbf{u}}_i\,|\Omega_i|$, reducing the continuous problem to a coupled system of algebraic equations for the cell-averaged unknowns, which is then advanced in time by a suitable explicit or implicit scheme [3].

2.2 Adaptive Mesh Refinement (AMR)

AMR is dynamic, so your explanation must focus on the rules that govern how the mesh changes during the simulation, particularly as the vortices travel downstream.

- *Refinement Criteria: This is the most critical detail. State the exact physical or numerical thresholds that trigger a cell to divide or merge. For a vortex street, this is typically driven by vorticity gradients or velocity curl.*
- *Resolution Limits: Define the maximum and minimum allowed grid sizes ($\Delta x_{min}, \Delta x_{max}$). You must also mention how the time step (Δt) adapts to maintain numerical stability, often referencing the Courant-Friedrichs-Lewy (CFL) condition where the Courant number $Co = \frac{u\Delta t}{\Delta x}$ is kept below a specific threshold.*
- *Computational Architecture: Dynamically changing meshes require rigorous memory allocation. If you built the simulation engine in a language like C, explain your data structures. Detail how you utilize pointers to manage the dynamically shifting arrays of the adaptive mesh, ensuring efficient handling of the large experimental dataset without constantly copying data across the CPU.*
- *Data Pipeline for Animation (HDF5): Briefly explain how the non-uniform data from the adaptive mesh is interpolated or extracted for the final video. If you use custom Python scripts for regression models or to parse the raw numerical output into plotting libraries, state this workflow.*

3 Results, animations and analysis

- *A brief paragraph in which the baseline models are described.*
- *Explanation of which parameters are changed based on the initial model, and how those changes are made.*

3.1 Qualitative analysis

The purpose of this section is to describe the visual morphology changes of all the fluids that are simulated. This is done by comparing different groups of animations at the same time, capturing this with one unique frame that represents the behaviour of the whole phenomenon in that moment. And those instants are not choose randomly, they are defined in the particular times in which some relevant event happen in the simulation.

3.1.1 Laminar incoming flow

3.1.2 Turbulent incoming flow

3.2 Quantitative discussion

To measure the behaviours that are easily described by visual descriptions, which are detailed in the previous sections, the following variable is defined.

$$r = < u^2 > \tag{5}$$

This parameter allows to define a generalization of the velocity field's behaviour. The idea with this magnitude is to define a certain convenient domain, which is fixed for all the simulations, and compare it between the different models. Using this it is possible to now make a proper quantitative analysis. And, based on r other numerical parameters can be designed to define certain patterns over the models established.

4 Conclusions

- *Problems with the simulations and their respective recommendations.*

- *Objectives accomplished?*
- *Results in comparison and contrast with the bibliography*

It is important to mention that the majority of these combinations of parameters are very unlikely to be useful for a particular field within the fluid dynamics domain. Since, the general goal to accomplish with this work is to characterize how the variations in the main variables of a fluid may change its behaviour. And, this is done based on the common phenomenon of a Von Kármán Vortex street. But the modifications that are made don't follow, in general, any guideline of a certain subdiscipline in the fluid area. So, the results are expected to not be suitable to any application, specially to experimental purposes. But instead, the predicted outcome is expected to contribute to a deeper understanding of the universal behaviour of fluids. Therefore, this objective is considered achieved.

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